Question 1 | Correct Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:
The first line contains the no of elements in the list-n
The next n lines contain the elements.

Output:
Sorted list of elements

For example:

Input	Result
5 67 34 12 98 78	12 34 67 78 98

Answer:

```c
#include <stdio.h>

void quick(int a[], int l, int h)
{
    if (l >= h) return;

    int i=l, j=h, p=a[l], t;

    while(i<j)
    {
        while(a[i]<=p && i<h) i++;
        while(a[j]>p) j--;

        if(i<j)
        {
            t=a[i]; a[i]=a[j]; a[j]=t;
        }
    }

    t=a[l]; a[l]=a[j]; a[j]=t;

    quick(a,l,j-1);
    quick(a,j+1,h);
}

int main()
{
    int n,a[100],i;
    scanf("%d",&n);

    for(i=0;i<n;i++) scanf("%d",&a[i]);

    quick(a,0,n-1);

    for(i=0;i<n;i++) printf("%d ",a[i]);
}
```

	Input	Expected	Got	
✔	5 67 34 12 98 78	12 34 67 78 98	12 34 67 78 98	✔
✔	10 1 56 78 90 32 56 11 10 90 114	1 10 11 32 56 56 78 90 90 114	1 10 11 32 56 56 78 90 90 114	✔
✔	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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